Databricks production planning

Summarize this article for me

This section provides a structured, phase-by-phase approach to planning and designing a production-ready enterprise Azure Databricks platform. It focuses on architectural decisions, design patterns, and best practices rather than step-by-step implementation instructions.

Overview

This section helps administrators understand core principles and design patterns for planning Azure Databricks account and production workspace deployments.

Who this is for

This section is designed for **enterprise production deployments** with complex governance, security, and multi-workspace requirements:

- Cloud architects designing enterprise Azure Databricks deployments.
- Platform engineers planning production Databricks infrastructure.
- Data architects designing governance and storage strategies for multiple teams.
- Security teams evaluating Azure Databricks security patterns for regulated environments.
- Account administrators deploying production workspace fleets.

Getting started instead? If you're new to Azure Databricks or exploring the platform, start by creating a serverless workspace. See Create a serverless workspace. You can return to this section when you're ready to design your production architecture.

What's covered

This section focuses on **design and architecture decisions**. Each phase presents design patterns, best practices, and strategic considerations. For step-by-step implementation instructions, refer to the documentation linked at the end of each phase.

Databricks Well-Architected Framework

Each phase includes best practices aligned with the Databricks Well-Architected Framework. For comprehensive architectural principles, see Databricks well-architected framework.

Prerequisites

Before beginning production planning, ensure you have:

- **Cloud account**: Active cloud account with appropriate admin permissions.
- **Azure Databricks account**: Account admin access to Azure Databricks account console.
- **Requirements gathering**: Understanding of your organization's security, compliance, and governance requirements.
- **Network planning**: Network architecture plan including CIDR ranges and connectivity requirements.
- **Identity provider**: Identity provider details for SSO integration (recommended for production).

Planning phases

This section consists of 10 phases. Phases can overlap or be executed in parallel depending on your organization's needs and existing infrastructure.

Phase execution strategies

- **Sequential**: Complete phases in order for greenfield deployments.
- **Parallel**: Execute independent phases simultaneously (for example, network and identity setup).

- **Iterative**: Revisit phases as requirements evolve (for example, add workspaces, expand to new regions).

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Phase	Description
Phase 1: Account	Configure foundational account administration and identity management strategy.
Phase 2: Workspace strategy	Plan workspace architecture based on organizational structure, security requirements, and operational needs.
Phase 3: Unity Catalog	Design Unity Catalog governance architecture including metastore patterns, catalog structure, and access control models.
Phase 4: Network	Design cloud network infrastructure to support Azure Databricks compute and data plane connectivity.
Phase 5: Storage	Design storage strategy for workspace storage and data storage across clouds.
Phase 6: Delta Lake	Design Delta Lake storage architecture and data organization patterns for Databricks.
Phase 7: IaC	Design IaC strategy to automate deployment and management of Azure Databricks resources.
Phase 8: Compute	Design compute strategy and workspace settings to optimize performance, cost, and security.
Phase 9: Observability	Design observability and monitoring strategies to ensure operational excellence.
Phase 10: High availability & DR	Design HA and DR strategies to ensure business continuity and resilience.

From design to implementation

After completing the design phases, implement your architecture using:

Infrastructure deployment

- Use Terraform to deploy account-level infrastructure (for example, workspaces, networks, Unity Catalog metastores).

- Use Declarative Automation Bundles to deploy data and AI workloads (for example, jobs, pipelines, notebooks, models).
- Automate deployments through CI/CD pipelines.

Validation and testing

- Test workspace connectivity and compute provisioning.
- Validate Unity Catalog permissions and data access patterns.
- Test network connectivity to data sources.
- Verify observability dashboards and alerts.

Additional resources

Documentation

- Databricks Well-Architected Framework
- Unity Catalog best practices
- Security and compliance

Next steps

Begin your production planning with Phase 1: Account.

Feedback

Was this page helpful? Yes No

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